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AS SKYROOT AEROSPACE'S diminutive 546 kg, 8-metre tall sub-orbital launch vehicle, Vikram-S, zoomed towards the exosphere on November 18, a round of applause erupted in the sounding rocket complex at the Satish Dhawan Space Centre (SDSC) in Sriharikota. Elsewhere in India, the founders of more than 100 spacetechn start-ups watching the event also celebrated the launch of the country's first privately-built rocket.

Within a week of the launch, the Polar Satellite Launch Vehicle of the Indian Space Research Organisation (ISRO) successfully deployed Anand and Thybolt 1 & 2—satellites of spacetechn start-ups Pixxel and Dhruva Space, respectively—in space.

The blistering pace of launches for Indian spacetechn start-ups was set on July 18 when high-end tech payloads of Bengaluru-based Digantara Research & Technologies and Hyderabad-based Dhruva were successfully launched from SDSC. Meanwhile, Skyroot is preparing to put its first orbital satellite launch vehicle, Vikram-1, into space within a year. "It will launch multiple satellites," says Pawan Kumar Chandana, Co-founder & CEO of Skyroot. "Over 50 per cent of the payload capacity is already booked," adds Naga Bharath Daka, Co-founder & CTO.

Anirudh Sharma, Co-founder & CEO of Digantara, says 2022 was "transformational" for the spacetechn segment. Within two years of India opening up its space sector to private players, several successes from spacetechn start-ups are already visible. "There has been a paradigm shift in the Indian private space ecosystem. Start-ups have repeatedly showcased their ability to build critical tech independently, and the government support has given them the confidence to build products for the world," says Sharma. The start-up is already building a space situational awareness observatory in



WHO'S DOING WHAT

SKYROOT AEROSPACE

► Small-lift, expendable Vikram series of launch vehicles designed to carry small satellites into Earth's orbit

AGNIKUL COSMOS

► Development of small-lift mobile launch systems such as Agnibaan to place 100-kg payloads in a 700-km orbit

PIXXEL

► Building a health monitor for Earth through a constellation of cutting-edge hyperspectral small satellites

BELLATRIX AEROSPACE

► Working on the development of in-space propulsion systems; also plans to launch its own two-stage rocket in 2023

DHRUVA SPACE

► Launching nanosatellites as well as developing satellites weighing up to 100 kg

GALAXEYE SPACE

► Development of some of the world's first satellites containing both Synthetic Aperture Radar (SAR) and optical sensors

DIGANTARA RESEARCH & TECHNOLOGIES

► Surveillance platform utilising a constellation of nanosatellites to provide predictive and situational awareness services

AZISTA-BST AEROSPACE

► Developing a facility in Ahmedabad to manufacture 250 satellites annually in a JV with Berlin Space Technologies